COMP6252 Deep Learning Technologies

Welcome!

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Department of Electronics and Computer Science

UNIVERSITY OF Southampton
School of Electronics and Computer Science

Lecturers

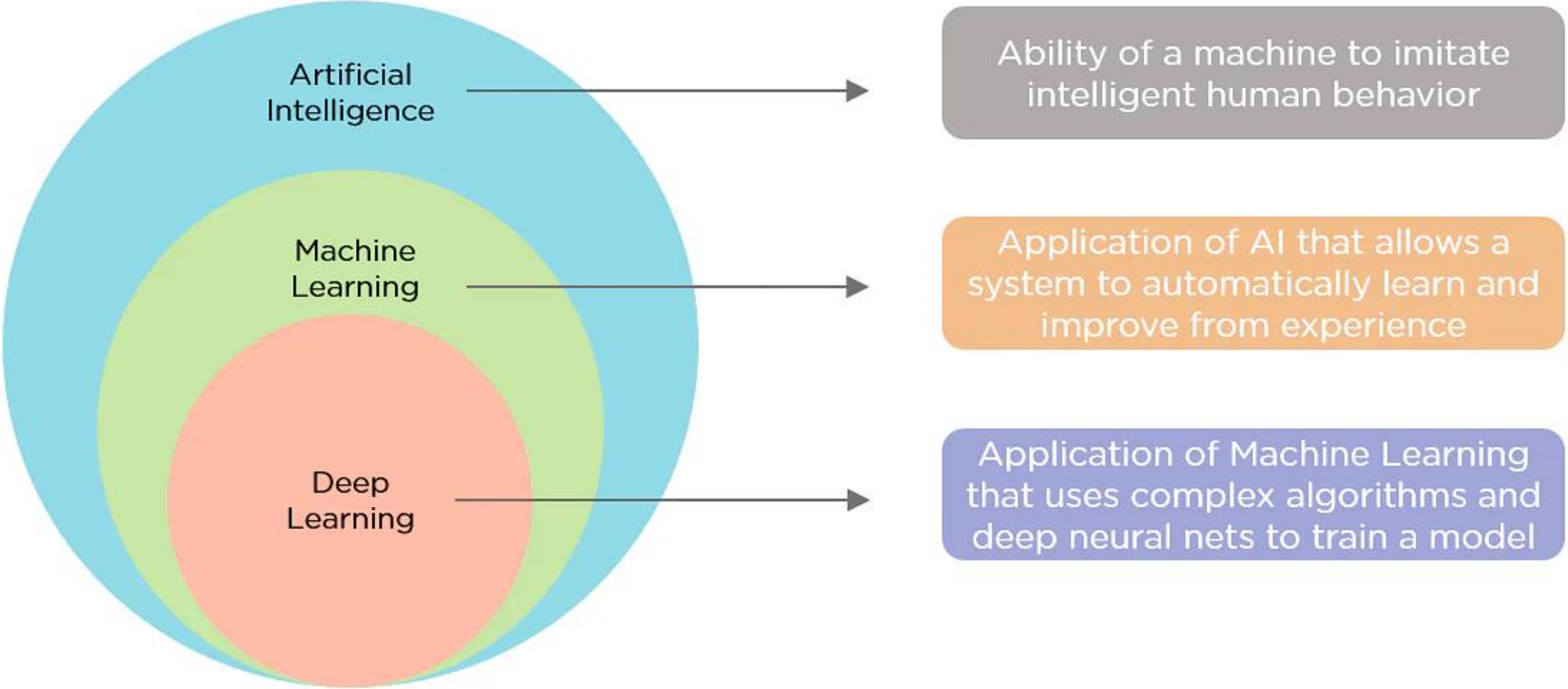
Hikmat Farhat

Zhiwu Huang

Xiaohao Cai

Welcome to Deep Learning Technologies

- It's a great subject
- Covers wide area
- We really enjoy it
- We shall try and impart the same to you!!



https://www.simplilearn.com/tutorials/artificial-intelligence-tutorial/ai-vs-machine-learning-vs-deep-learning

What will the course include?

- Live lectures

  Tuesday 12-13; Tuesday 13-14; and Friday 14-15

  …. which will also be recorded and placed on Panopto
- Live demonstrations in the lectures
- Plenty of time for Q&A
- … and bits we haven't thought of yet!

https://secure.ecs.soton.ac.uk/module/2526/COMP6252/43095/

Course website:

UNIVERSITY OF Southampton
School of Electronics and Computer Science

University of Southampton > ECS > Intranet > Modules > 2025-2026 > **COMP6252**

COMP6252: Deep Learning Technologies (2025-2026)

Overview | Resources | Syllabus | Evaluation | Send Message | Students | Help — You are a leader on this module.

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Module Leader

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Lecturer

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Lecturer

Jonathon Hare - **jsh2@soton.ac.uk**
Moderator

Southampton campuses, Semester 2.

Contents

Welcome to the COMP6252 "Deep Learning Technologies" 2024-25!

Deep learning has revolutionised numerous fields in recent years. We've witnessed improvements in everything from image processing and computer vision through speech analysis to natural language processing as a result of the advent of cheap GPGPU compute coupled with large datasets and some neat algorithms.

This module will look at how deep learning works in practical implementation, with the main focus on technologies. We'll look at models for deep learning such as convolutional and recurrent neural networks, as well as considering current research in depth.

Deep learning applications are almost limitless

- Feature extraction
- Image classification/generation
- Object detection
- Speech recognition
- Text recognition/generation
- Fraud detection
- …

Image captioning

man in black shirt is playing guitar.

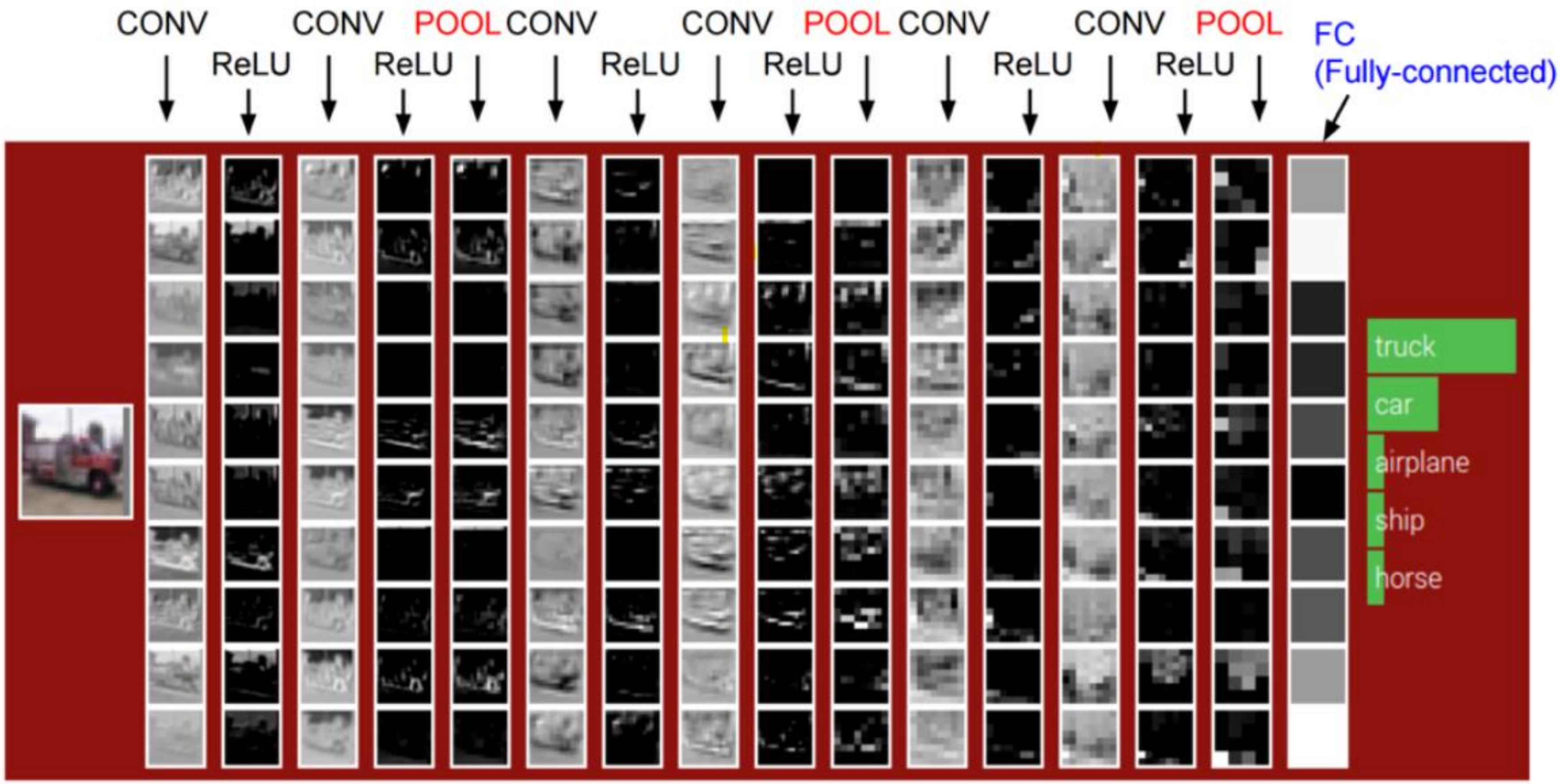
construction worker in orange safety vest is working on road.

two young girls are playing with lego toy.

boy is doing backflip on wakeboard.

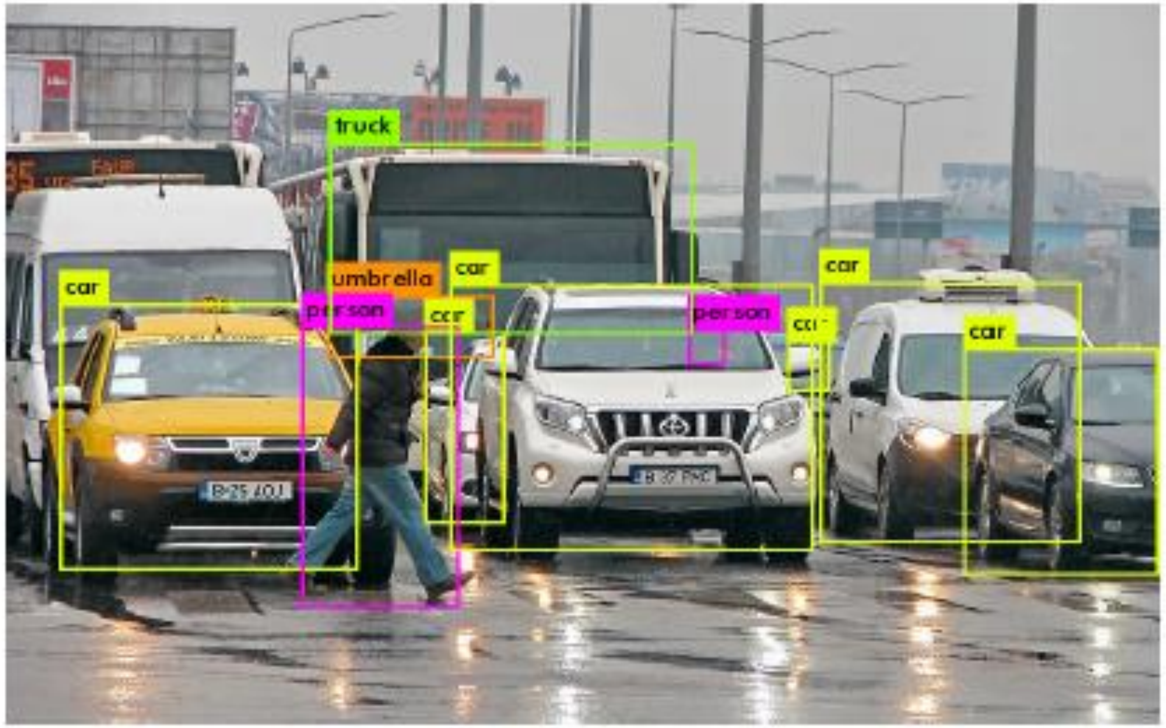
Karpathy and Fei-Fei 2015

Image classification



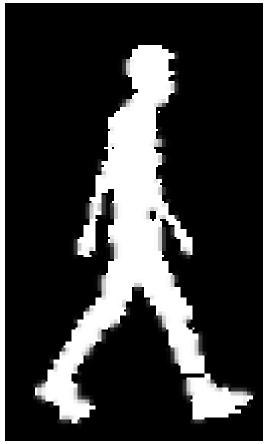
Andrej Karpathy

Object detection



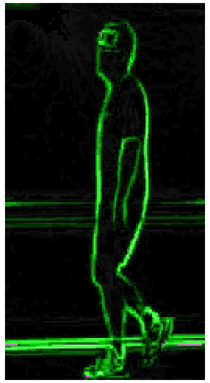
Gait Recognition

Recognising people from the motion of the whole body



silhouette

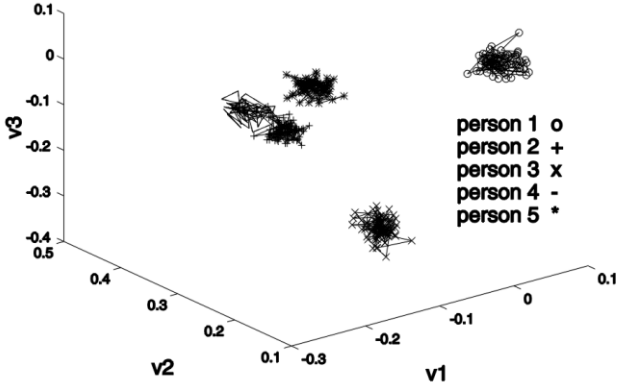
flow



edges

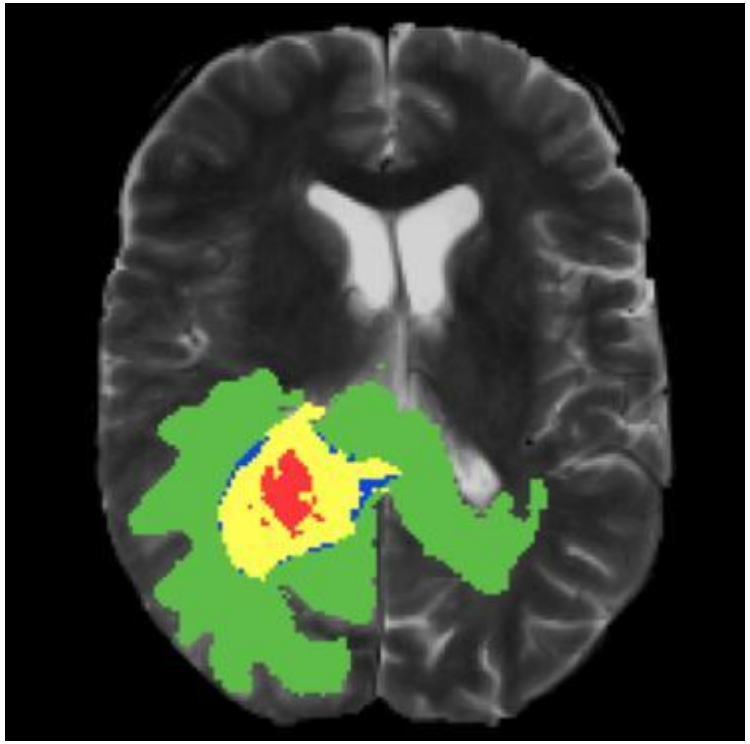
symmetry

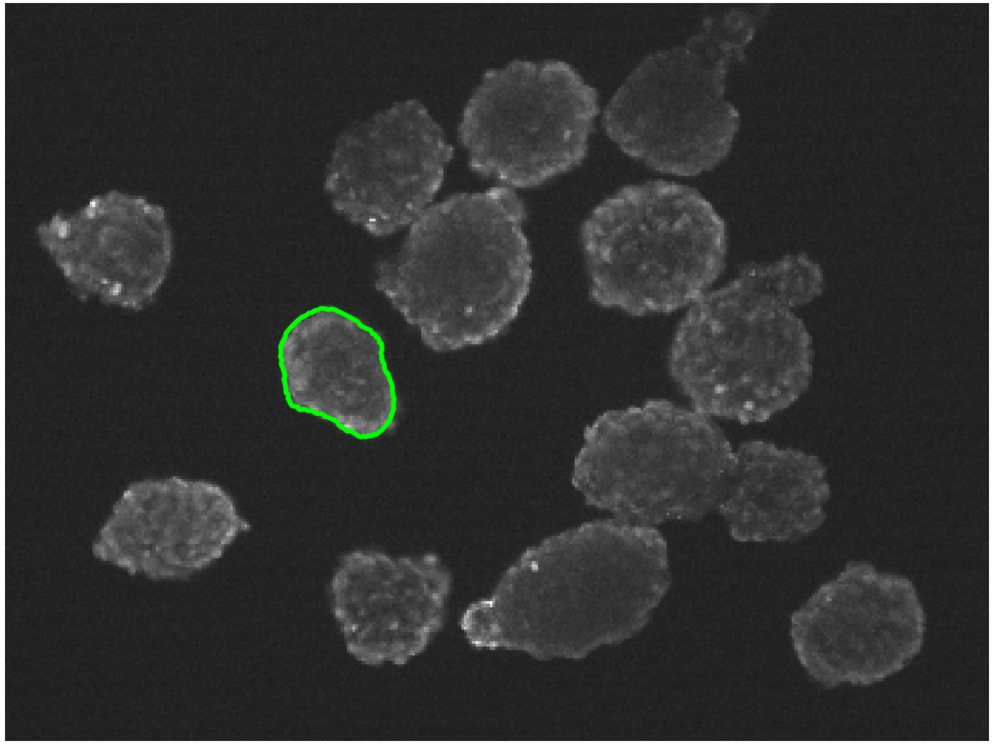
acceleration



feature space

Medical imaging





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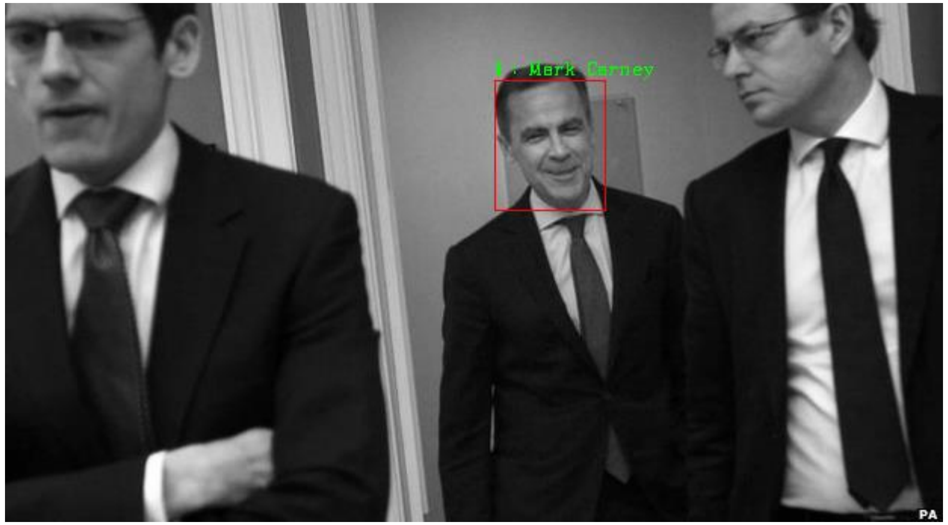
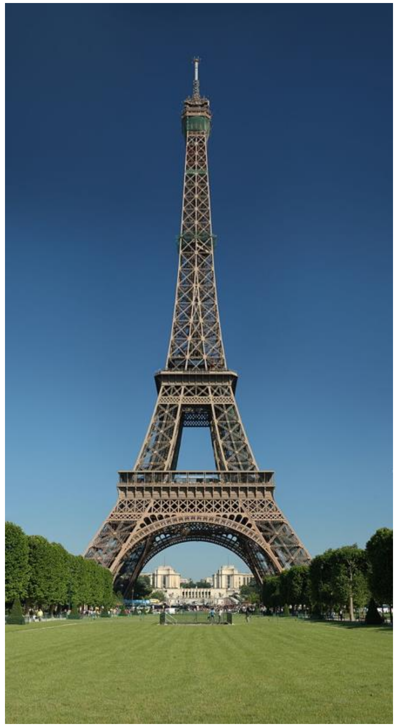
Higher level visual cognition

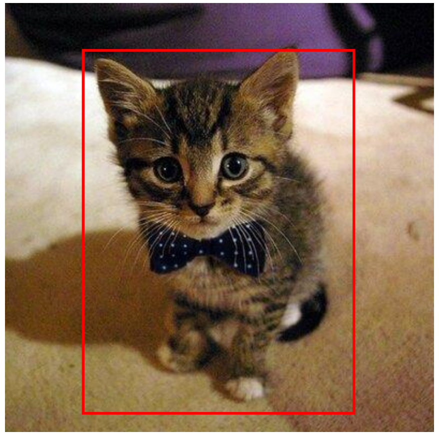


Who?



Where?

What?



Why?

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Deep learning support

- WWW homepages

  https://secure.ecs.soton.ac.uk/module/2526/COMP6252/43095/

  Lecture support materials

- Links
- Notes
- Extra Tutorials (on demand – we should have time near the end)
- Labs

COMP6252 - Deep Learning Technologies

PUBLISHED This list is live, students can see latest changes. ✓ Changes saved

Publish

ACADEMIC YEAR 2024/25

By Xiaohao Cai

Created 8 months ago | Updated 6 months ago

Linked to COMP6252

34 students

Edit View & Export My Lists

Description: Readling list for Deep Learning Technologies

Table of Contents | Type: All | Filter: All | Citation Style: None | Search

Deep Learning with Pytorch Quick Start Guide: Learn to Train and Deploy Neural Network Models in Python
Book - by David Julian - 2018 - Essential
VIEW ONLINE

Deep Learning
Website - by Ian Goodfellow - Essential
https://www.deeplearningbook.org/
VIEW ONLINE

Modern Computer Vision with PyTorch
Book - by V Ayyadevara; Yeshwanth Reddy; O'Reilly for Higher Education (Firm); Safari, an O'Reilly Media Company - 2020 - 1st edition - Additional
VIEW ONLINE

Neural networks and deep learning: a textbook
Book - by Charu C. Aggarwal - 2018 - Additional
VIEW ONLINE

Pattern recognition and machine learning
Book - by Christopher M. Bishop - ©2006 - Additional
VIEW AVAILABILITY

Hands-on machine learning with Scikit-Learn, Keras and TensorFlow: concepts, tools, and techniques to build intelligent systems
Book - by Aurélien Géron - 2023 - Third edition - Additional
VIEW ONLINE

Assessment

- Mixture of coursework and final ‘exam’
- 50% exam; 50% coursework

- Two courseworks (total 50%); designed to support learning:
  - 1 individual; 20%
  - 1 in groups of 4; 30%

Coursework schedule

Coursework

The schedule for each coursework is shown below, and details and specification will become available as each coursework is set. Deadline dates are currently provisional as deadlines might be moved by the Director of Programmes to minimise conflicts with other assignments from different modules.

Topic	Set	Due	Feedback
1. Coursework 1: Music Genre Classification (20%)	Tues 3rd Feb	16:00 Tues 28th Apr	Fri 8th May
3. Coursework 2: Application of Deep Learning Techniques (Group work, 30%)	Tues 10th Mar	16:00 Tues 5th May	Fri 15th May

Lecture Timetable

This course has about 20 lectures of stuff

- Hikmat will lead next for about 5 weeks [14 lectures]
- Xiaohao will then take over the lead for 1 week
- Zhiwu will then take over the lead for 1 week

Labs

- Time to be arranged
- Running for 9 weeks starting from next week
- Where: **Microsoft Teams**
- **You need to attend >=2 in order to complete your courseworks**

Labs content: Panel Discussion on Hot Deep Learning Topics, Lecture, Coursework Help and Advice!

Caveat: this module is also challenging!

- It contains 2 courseworks and final closed book exam.
- This module will just introduce popular deep learning technologies.
- If you would like to go deeper including the theoretical aspects…

Please do not select this module!

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Finally

✓ **Enjoy!**

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